Exercise

You are given a tiny dataset of fruits characterized by two features:

ID	Weight (g)	Color (1=Red, 0=Green)	Class (Label)
1	150	1	Apple
2	170	1	Apple
3	140	0	Pear
4	130	0	Pear
5	180	1	Apple

You must classify a **new fruit**:

Fruit X → Weight = 145 g, Color = 0 (Green)

Part 1 — Binary Decision Tree

1. Using **Weight** and **Color**, manually construct a *binary decision tree* that separates Apples and Pears.
    - Choose a split threshold for **Weight** (e.g., 160 g).
    - Then, optionally split again using **Color** if needed.
2. Draw the tree structure (by hand).
3. Using the tree, classify the new fruit (145 g, Green).

Questions:

- What is the root feature and its threshold?
- Which leaf node does Fruit X fall into?
- What class is predicted?

Part 2 — k-Nearest Neighbors (k-NN)

1. Use the same dataset and compute **Euclidean distances** to Fruit X based on both features.

$$d = \sqrt{(Weight_X - Weight_i)^2 + (Color_X - Color_i)^2}$$

2. Rank the 5 training fruits by distance to Fruit X.
3. For $k = 3$:
    - Find the 3 closest fruits.
    - Predict the class by majority vote.

Questions:

- What are the 3 nearest neighbors?
- What is the predicted class?
- How would the result change if $k = 5$?

Part 3 — Random Forest

Now imagine you build **3 small decision trees**, each trained on a *bootstrap sample* of the dataset and allowed to split on one random feature at each node.

1. Sketch 3 slightly different trees (they can differ by root feature or thresholds).
2. Predict the class of Fruit X for each tree.
3. Use **majority voting** among the 3 trees to get the final Random Forest prediction.

Questions:

- What are the 3 individual tree predictions?
- What is the ensemble's final prediction?
- Why might Random Forests be more robust than a single tree?